AI Satellite Annotation & Crop Analytics Tool

Comprehensive Developer, Architectural & Code Explanation Guide

1. System Architecture & High-Level Pipeline

The system integrates Deep Learning (Meta's Segment Anything Model & ResNet-50) with Geospatial Information Systems (GIS) to create an advanced interactive platform for satellite image annotation and agricultural decision support.

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        │  4-Band GeoTIFF Satellite   │
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         ┌───────────────────────┐
         │ 1. geo_processor.py   │
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│ 2. sam__predectorr.py │           │  3. classifier.py    │
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         │ 4. projects_manager.py│
         └──────────────┬──────────┘
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     ┌────────────────────────┐
     | Unified metadata.json  |
     |   & COCO AI Format     |
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2. In-Depth Technical Code Explanations

Module 1: geo_processor.py (The Geospatial Analytics Engine)

This module acts as the entry point for handling 4-band satellite raster images (Red, Green, Blue, Near-Infrared). It handles structural decoding and calculates vegetative indices.

Key Functions and Internal Logic:

- **process_geotiff(file_bytes)**:
  - **RasterIO Memory Ingestion**: Uses rasterio.MemoryFile to open raw binary streams into active geospatial raster datasets without touching the physical disk, maximizing processing speed.
  - **Band Slicing**: Extracts Band 1 (Red), Band 2 (Green), Band 3 (Blue), and Band 4 (NIR) dynamically.
  - **Standard RGB Mapping**: Combines Red, Green, and Blue channels using np.dstack and scales dynamic reflectance values to standard 8-bit image pixels (0-255) via Min-Max normalization.
  - **NDVI Computation**: Computes the Normalized Difference Vegetation Index using the mathematical formula:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

  - **Advanced Dynamic Heatmap Masking**:
    - Pixels with $\text{NDVI} > 0.1$ are isolated as active healthy vegetation and mapped dynamically to a **Yellow-to-Green (YlGn)** colormap.

        - Pixels with $\text{NDVI} \le 0.1$ are classified as poor/bare soil and strictly masked to a fixed **Red/Brown** color [180, 40, 40], entirely preventing false vegetation readings.
    - **Decision Support Statistics (DSS)**: Calculates real-time total land covered, pixel ratios, mean NDVI value, and standard deviation to give instant asset reports.

Module 2: sam__predectorr.py (Interactive Segmentation Engine)

This module contains the AdvancedSAMSegmenter class, which handles zero-shot click-driven polygon segmentation on satellite patches.

Key Functions and Internal Logic:

- **set_image(file_bytes)**:
    - Loads the image, applies downscaling if dimensions exceed 2048px to guarantee stable GPU VRAM overhead, and feeds the embedding network.
    - Extracts spatial transformation matrices (src.transform) and Coordinate Reference Systems (src.crs) to maintain absolute spatial reference.
- **segment_with_click(click_x, click_y, click_type, class_id, color_hex)**:
    - **Thread Safety**: Uses a threading.Lock() mutex to handle asynchronous API user requests safely without server race conditions.
    - **Positive/Negative Prompting**: Translates standard UI clicks into positive foreground hints (click_type=1 to grab a crop parcel) or negative background hints (click_type=0 to refine and subtract wrong selections).
    - **Mask Merging**: Updates a global mask stack (self.combined_mask) using boolean array logic.
    - **Geospatial Georeferencing & Spatial Math**: Calculates pixel-to-ground area mapping using metadata resolutions. Translates pixel bounding boundaries into absolute geo-coordinates (minx, miny, maxx, maxy).
- **export_geojson(class_id)**:
    - Uses rasterio.features.shapes to extract exact vector contours (Polygons) from binary pixel arrays and formats them into a standard GeoJSON file structure.

Module 3: classifier.py (Deep Learning ResNet-50 Crop Identifier)

This class (EuroSATResNetClassifier) acts as the secondary validation phase by identifying crop specifics inside boundaries using a custom-trained network.

Key Functions and Internal Logic:

- **__init__(weights_path)**:
  - Constructs a deep convolutional backbone using a structural **ResNet-50** architecture.
  - **Fully Connected (FC) Layer Adaptation**: Overwrites self.model.fc from the original 1000-class ImageNet setup to target exactly **10 classes** to prevent dimension shape errors.
  - Loads your trained state dictionary (satellite_model.pth) and enforces standard eval() mode to fix BatchNorm and Dropout settings.
- **Standard Transforms**: Forces input inputs into dimensions of (224, 224) and applies standard normalization factors:

$$\text{Mean} = [0.485, 0.456, 0.406], \quad \text{Std} = [0.229, 0.224, 0.225]$$

- **predict_patch(cropped_rgb_array)**:
  - Converts pixel crops to PIL format, applies forward-pass tensors, disables gradient tracking via torch.no_grad() to save GPU memory, and applies Softmax to convert raw logits into clean confidence scores matching the **10 EuroSAT classes**.

Module 4: projects_manager.py (Data Integrity & Export Manager)

This file handles standard system I/O, updates the persistent state database, and exports datasets into international file types.

Key Functions and Internal Logic:

- **create_project() & get_projects()**: Programmatically creates directories for isolated user operations under a central path (data_projects/).
- **update_project_statistics(project_name, class_id, stats_data)**:
  - Reads the structural project database, injects the real-time calculated area, spatial coordinates, predicted ResNet labels, and confidence metrics, then commits it back to disk using indented json.dump().

- **export_to_coco_format(...)**:
    - Transforms local metadata schemas into the internationally accepted **COCO JSON Format**.
    - Structures the resulting JSON file with the mandatory root arrays: info, licenses, images, categories, and annotations.
    - Maps polygon pixel indices and bounding dimensions into standard pixel arrays [x, y, width, height], transforming your app into an official **AI Dataset Generator**.

Module 5: app.py & app_integrated.py (The Portal Interconnection)

This file sets up a responsive UI built on top of **Streamlit**. It hooks mouse-click actions on standard map components, pulls processed arrays from geo_processor, passes coordinate vectors down to sam__predectorr, runs classifier, and logs outputs through projects_manager smoothly.

3. The Runtime Pipeline (What Happens on a Single User Click)

[User Click Event: X, Y]

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[SAM Engine processes prompt] ──► Extracts boundaries, maps Geo-coordinates, and computes total area.

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[BBox Bounding Box Extraction] ──► Crops the target field from the original 4-band raw canvas.

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[ResNet-50 Classification] ──► Matches pixels against 10 EuroSAT classes and assigns a confidence score.

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[Projects Manager Update] —► Encapsulates all data and appends it to metadata.json & COCO JSON on disk.

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[Real-Time UI Refresh] —► Updates the interactive map overlay and refreshes the analytics dashboard.

4. Engineering Comparison: The Power of COCO Export Feature

Technical Metric	**Legacy Custom Format (Without COCO)**	**Standardized COCO AI Format (With COCO)**
Data Structure	Custom-designed layout for UI rendering.	Global universal format (info, images, annotations).
Model Reusability	Incompatible with standard model pipelines without conversion.	Direct import capability into models like YOLO, Mask R-CNN, and Faster R-CNN.
Coordinate System	Relies on geographical coordinates.	Translates data into precise pixel bounds for direct model training.
Academic Value	Viewed as a basic mapping application.	Recognized as an official **AI Dataset Generator**, significantly boosting academic and practical value.